



BIO 004 • HUMAN ANATOMY • SOLANO COMMUNITY COLLEGE

Module 1 Lab Structure List.

Orientation • Tissues • Integumentary

Contents

1. Introduction to Anatomy	3
Landmarks, orientation, cavities	
2. Epithelial Tissues	6
Slides and models	
3. Connective Tissues	7
Proper, cartilage, osseous	
4. Muscle Tissues	9
Skeletal, cardiac, smooth	
5. Nerve Tissues	9
Tested in Module 5	
6. Blood Vessels and Formed Elements	10
Tested in Module 3	
7. Integumentary System	11
Epidermis, dermis, hypodermis	

How to use this list

This is a quick reference lab study guide. These are the lab structures and concepts to learn for the lab portion of the exam. Refer to the lab worksheets for the study questions that matter for the course. Treat this as the master list. Occasionally a lab sheet will not contain a structure seen here, so always cross reference against this list. Be prepared to identify structures on models, images, the cadaver, and slides from the microscope.

Some items are tagged with the module they are tested in. You meet the slide in Module 1 and are examined on it later.

Each tissue gets its own block. Tick the box by the name once you can identify it on the slide, write in its function and location, then work down the vertical list and tick each structure you can point to. Anything not in a list is context, not something you will be asked to identify.

READ THIS PART TWICE

This is an extensive list. It would be impossible to test on everything, but everything on this list is testable and you should prepare for it. On occasion I may add something to the list or remove something. Make sure you are in class, and have a person to connect with if you miss class, who can update you on any changes.

CHAPTER 1

Introduction to Anatomy

Body landmarks, orientation, planes, cavities, membranes, and the microscope. Everything here is testable on models, images, the cadaver, and slides.

1. **All anterior and posterior body regional landmarks.** Know the term, and be able to point to it on a model and on yourself.

Anterior regions

ANTERIOR REGIONAL TERMS

Term	Region	Term	Region
Frontal	Forehead	Inguinal	Groin
Cranial	Skull or head	Pubic	Area near the genital region
Facial	Face	Brachial	Upper arm
Orbital or ocular	Eye region	Antebrachial	Forearm
Buccal	Cheek	Antecubital	Front of the elbow
Otic	Ear region	Carpal	Wrist
Nasal	Nose	Pollex	Thumb

Term	Region	Term	Region
Oral	Mouth region	Palmar	Palm of the hand
Mental	Chin	Digital or phalangeal	Fingers or toes
Cervical	Neck region	Patellar	Kneecap
Thoracic	Chest area	Crural	Leg, the shin area
Sternal	Breastbone	Tarsal	Ankle
Mammary	Breast region	Pedal	Foot
Abdominal	Below the chest, above the pelvis	Hallux	Big toe
Umbilical	Navel or belly button	Axillary	Armpit
Pelvic	Lower abdomen or hip area	Coxal	Hip
Femoral	Thigh		

Posterior regions

POSTERIOR REGIONAL TERMS			
Term	Region	Term	Region
Cephalic	Head	Sacral	Area between the hips, over the sacrum
Occipital	Back of the head	Gluteal	Buttock region
Cervical	Neck	Femoral	Thigh
Acromial	Point of the shoulder	Popliteal	Back of the knee
Scapular	Shoulder blade	Sural	Calf
Dorsal	Back	Fibular or peroneal	Lateral side of the leg
Vertebral	Spinal column	Calcaneal	Heel
Olecranal	Back of the elbow	Plantar	Sole of the foot
Lumbar	Lower back or loin area		

2. **Identify those structures in terms of body orientation and direction.** Be able to use each in a sentence. The trachea is ventral to the spine. The nose is medial to the cheekbone.

DIRECTIONAL TERMS, IN PAIRS

Term	Its opposite	What the pair describes
Anterior (ventral)	Posterior (dorsal)	Toward the front, versus toward the back
Superior	Inferior	Above, versus below
Medial	Lateral	Toward the midline, versus away from it
Cephalad (cranial)	Caudal	Toward the head, versus toward the tail end
Proximal	Distal	Nearer the trunk, versus farther from it. Limbs only
Superficial	Deep	Toward the body surface, versus away from it
Ipsilateral	Contralateral	On the same side of the body, versus the opposite side

3. **Body planes and sections.** Midline, median (midsagittal), parasagittal, coronal (frontal), transverse (horizontal or cross-sectional), oblique.
4. **Body cavities.** Ventral and dorsal, superior and inferior, cranial, vertebral, thoracic, abdominopelvic, mediastinum.

What is the name of the structure that separates the thoracic cavity from the abdominopelvic cavity?

5. **Mediastinum.**

1. Know the structures that form the following borders: superior, inferior, ventral, dorsal, and lateral.

2. Know the contents of the mediastinum.

6. **Retroperitoneum.** Name the retroperitoneal structures. Memory device: SADPUCKER.

7. **Serous membranes and cavities.** Pericardial (pericardium): parietal, visceral, cavity, fluid. Peritoneal (peritoneum): parietal, visceral, cavity, fluid. Pleural (pleura): parietal, visceral, cavity, fluid.

8. **Abdominopelvic quadrants and the major structures found within each.**

THE FOUR QUADRANTS AND THEIR ORGANS

Quadrant	Organs found within
Right upper (RUQ)	Liver (right lobe), gallbladder, right kidney, duodenum, head of the pancreas, hepatic flexure of the colon, part of the ascending and transverse colon
Left upper (LUQ)	Stomach, spleen, left lobe of the liver, left kidney, body and tail of the pancreas, splenic flexure of the colon, part of the transverse and descending colon
Right lower (RLQ)	Cecum, appendix, ascending colon, right ureter, right ovary and uterine tube, right spermatic cord
Left lower (LLQ)	Descending colon, sigmoid colon, left ureter, left ovary and uterine tube, left spermatic cord

The urinary bladder and the uterus sit at the midline in the hypogastric region and can be described in either lower quadrant.

9. Be able to calculate total magnification, and name the parts of the microscope.

CHAPTER 2

Epithelial Tissues

Identify the tissue on the slide, then write its function and location. Tick each structure as you find it.

General structures, on every epithelial slide

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Apical surface | ☐ Lamina propria, when present | ☐ Microvilli, when present |
| ☐ Lateral surface | ☐ Nuclei | ☐ Goblet cells, when present |
| ☐ Basal surface | ☐ Lumen or free surface | ☐ Keratin, when present |
| ☐ Basement membrane | ☐ Cilia, when present | |

☐ Simple squamous

FUNCTION _____ LOCATION _____

☐ Keratinized stratified squamous

FUNCTION _____ LOCATION _____

☐ Non-keratinized stratified squamous

FUNCTION _____ LOCATION _____

☐ Simple cuboidal

FUNCTION _____ LOCATION _____

☐ **Stratified cuboidal**

FUNCTION _____ LOCATION _____

☐ **Simple columnar**

FUNCTION _____ LOCATION _____

☐ **Pseudostratified ciliated columnar**

FUNCTION _____ LOCATION _____

☐ **Transitional**

FUNCTION _____ LOCATION _____

Which tissues typically contain cilia? _____ Function: _____

Microvilli? _____ Function: _____

Goblet cells? _____ Function: _____

CHAPTER 3**Connective Tissues**

Connective tissue proper first, then cartilage and bone. For Module 1 you need the basic identification. The structures inside cartilage and bone come back in Module 2.

Connective tissue proper☐ **Loose areolar**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY☐ Collagen fibers☐ Fibroblasts☐ Elastic fibers☐ Mast cells☐ **Adipose**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY☐ Adipocytes☐ Fat vacuoles containing lipids☐ Blood vessels☐ **Reticular**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY☐ Reticular fibers

☐ **Dense regular**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Collagen fibers ☐ Fibroblasts☐ **Dense irregular**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Collagen fibers ☐ Fibroblasts☐ **Elastic**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Elastin fibers**Cartilage and osseous tissue** BASIC ID IN MODULE 1 STRUCTURES IN MODULE 2

Name the tissue, say what it does and where it is. Tick the structures you can already find, the rest is Module 2 work.

☐ **Hyaline cartilage**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Perichondrium ☐ Chondrocytes
☐ Lacunae ☐ Extracellular matrix☐ **Elastic cartilage**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Perichondrium ☐ Chondrocytes ☐ Elastin fibers
☐ Lacunae ☐ Extracellular matrix☐ **Fibrocartilage**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Lacunae ☐ Extracellular matrix ☐ No perichondrium
☐ Chondrocytes ☐ Collagen fibers☐ **Osseous tissue**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Osteons ☐ Lacunae ☐ Osteocytes
☐ Central (Haversian) canal ☐ Lamellae ☐ Canaliculi**CHAPTER 4**

Muscle Tissues

Basic identification for Module 1. Name the tissue, what it does, where it is, and whether it is voluntary or involuntary. The sarcomere and the connective tissue wrappings come back in Module 3.

☐ Skeletal muscle

FUNCTION _____ LOCATION _____

VOLUNTARY OR INVOLUNTARY _____

STRUCTURES TO IDENTIFY

- ☐ Striations, dark and light ☐ Nuclei, multiple and peripheral ☐ Sarcolemma

☐ Cardiac muscle

FUNCTION _____ LOCATION _____

VOLUNTARY OR INVOLUNTARY _____

STRUCTURES TO IDENTIFY

- ☐ Branching fibers ☐ Single central nucleus
☐ Intercalated discs ☐ Striations

☐ Smooth muscle

FUNCTION _____ LOCATION _____

VOLUNTARY OR INVOLUNTARY _____

STRUCTURES TO IDENTIFY

- ☐ Spindle-shaped cells ☐ Single central nucleus ☐ No striations

What two cell junctions are found at intercalated discs, and what does each one do?

CHAPTER 5

Nerve Tissues

Basic identification for Module 1. The detail comes back in Module 5.

☐ Multipolar neuron

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

- ☐ Cell body (soma) ☐ Dendrites ☐ Synaptic bulb
☐ Nucleus ☐ Axon
☐ Nucleolus ☐ Axon hillock

☐ Spinal cord

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

- ☐ White matter ☐ Ventral horn ☐ Lateral horn, when present
☐ Grey matter ☐ Dorsal horn

☐ **Cerebral cortex**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Pyramidal cells☐ **Cerebellum**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Purkinje cells☐ **Nerve, cross section**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Axon☐ Nodes of Ranvier☐ Perineurium☐ Schwann cells☐ Epineurium☐ Endoneurium

CHAPTER 6

Blood Vessels and Formed Elements

Basic identification for Module 1. The detail comes back in Module 3.

Vessel walls☐ **Artery**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Tunica intima☐ Tunica media☐ Tunica externa☐ Endothelium☐ Smooth muscle☐ Lumen☐ Internal elastic lamina☐ Dense elastic connective tissue☐ Erythrocytes☐ **Vein**

FUNCTION _____ LOCATION _____

STRUCTURES TO IDENTIFY

☐ Tunica intima☐ Tunica media with smooth muscle☐ Lumen☐ Endothelium☐ Tunica externa☐ Erythrocytes**Peripheral blood smear**

Identify each formed element, then write its function and how you recognized it.

☐ **Neutrophil**

FUNCTION _____ LOCATION _____

☐ **Basophil**

FUNCTION _____ LOCATION _____

☐ **Eosinophil**

FUNCTION _____ LOCATION _____

☐ **Monocyte**

FUNCTION _____ LOCATION _____

☐ **Lymphocyte**

FUNCTION _____ LOCATION _____

☐ **Erythrocyte**

FUNCTION _____ LOCATION _____

☐ **Thrombocyte**

FUNCTION _____ LOCATION _____

CHAPTER 7

Integumentary System

Layers in order, deep to superficial, then the structures in the dermis and hypodermis.

Strata of the epidermis☐ **Stratum corneum**

FUNCTION _____ LOCATION _____

☐ **Stratum lucidum**

FUNCTION _____ LOCATION _____

☐ **Stratum granulosum**

FUNCTION _____ LOCATION _____

☐ **Stratum spinosum**

FUNCTION _____ LOCATION _____

☐ **Stratum basale**

FUNCTION _____

LOCATION _____

Dermis
☐ **Dermis, on slides and models**

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Papillary layer | ☐ Hair | ☐ Sebaceous gland and duct |
| ☐ Reticular layer | ☐ Hair follicle | ☐ Apocrine gland and duct |
| ☐ Areolar connective tissue | ☐ Hair root | ☐ Meissner's corpuscles |
| ☐ Dense irregular connective tissue | ☐ Hair bulb | ☐ Free nerve endings |
| ☐ Dermal papillae | ☐ Hair shaft | ☐ Capillary loops |
| ☐ Eccrine sweat gland and duct | ☐ Arrector pili muscle | |

Hypodermis, the subcutaneous tissue
☐ **Hypodermis**

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Adipose connective tissue | ☐ Areolar connective tissue | ☐ Lamellated (Pacinian) corpuscle |
|--|--|--|

Name the three structures found within dermal papillae.

Which epidermal layer is missing in thin skin? _____